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Adaptor-Based On GaN



The reference design enables efficient power conversion with a high-voltage converter and GaN transistor, supports USB power delivery, and includes protection features for safe operation across various electronics.

The EVLVIPGAN50PD reference design from STMicroelectronics provides engineers with a solution for developing USB Type-C power delivery 3.0 adaptors. It leverages the high-voltage converter with an integrated 650V HEMT power GaN transistor, enabling the design of isolated power supplies optimised for quasi-resonant flyback converters, delivering up to 50W across a wide input range. Including the SRK1001 adaptive synchronous rectification controller improves efficiency on the secondary side, ensuring power conversion. The design also integrates the STUSB4761 controller for CC/CV regulation and USB Type-C PD interface management, enabling a hardware-based USB power delivery stack that ensures safe and compliant device communication. Built-in protection features guard against overvoltage, undervoltage, overpower, and short circuits, making it suitable for various electronics.

The reference design can be used to create advanced USB Type-C power delivery 3.0 adaptors for various applications.

STMicroelectronics

<https://www.electronicshobby.com/electronics-projects/adaptor-reference-design-based-on-gan>

65W GaN-Based Adaptor



The reference design integrates GaN technology for higher switching frequencies, reduced losses, and improved efficiency. Its design supports USB power applications, with output voltage adjustments and lossless current sensing for optimised performance.

TIDA-050072, reference design from Texas Instruments integrates gallium nitride (GaN) technology for higher switching frequencies, reduced losses, and improved efficiency, while a quasi-resonant flyback topology optimises efficiency by adjusting the switching frequency to minimise drain voltage on the primary FET. The design includes a GaN device with sense emulation for lossless current sensing, enhancing system efficiency. GaN technology allows for smaller inductors, reduced core sizes, and fewer primary windings, making the design suitable for mobile wall chargers, USB power outlets, and auxiliary power supplies. Engineers can optimise transformer design for high-frequency operation, ensuring design efficiency while selecting switching devices to manage frequency- and current-dependent conduction losses. The board includes a USB Type-C PD controller that connects to a USB Type-C PD load through an onboard connector, allowing output voltage adjustments to 5V, 9V, 15V, or 20V.

The design helps engineers create power solutions for charging devices, including mobile phones, laptops, and tablets.

Texas Instruments (TI)

<https://www.electronicshobby.com/electronics-projects/65-w-gan-based-adaptor-reference-design>

GaN USB PD Dual-C Charger



The reference design supports multi-device charging with power sharing, efficiency, and a wide input range. Its form factor and power density enable chargers to meet energy standards without performance loss.

Infineon's REF_65W_2C_GAN_PAG1 reference design enables engineers to develop multi-device chargers that comply with energy efficiency standards like DOE Level VI. Its form factor and power density allow for small, portable chargers without sacrificing performance. Designed for USB-PD multi-port applications, it supports a wide input range with output voltages from 3.3V to 21V. Each USB-C port delivers 65W, with a total output of 65W dynamically shared based on connected devices. The QR flyback charger adaptor achieves over 93% efficiency and allows for customisation. The design includes an ATQ transformer and integrates two controllers for load sharing. It operates within a 90-264V AC input range at 50-60Hz. The board measures 56mm×27mm×33mm, achieving a power density of 21.34W/in³, with efficiency over 92% at full load.

The reference design is suitable for USB-PD multi-port charger applications.

Infineon

<https://www.electronicshobby.com/electronics-projects/gan-usb-pd-dual-c-charger-reference-design>